Mohsen Pirmoradiyan

AI Lead & Computer Vision Engineer

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PROFILE

Computer-vision and AI engineer who builds production deep-learning systems for drone and aerial imagery: segmentation, self-supervised learning and geospatial ML. Before moving into machine learning I spent fourteen years in seismic field operations, which shaped how I work: start from the signal, check the data before trusting it, and think in geometry. The full story, from the oil fields to computer vision, with a page for each step of my career and my learning notes, is on my website: mpirmoradiyan.github.io.

EXPERIENCE

Feb 2022 – present

AI Lead & Computer Vision Engineer

Arboair AB, Linköping, Sweden · drone and aerial analytics for forestry

- Designed and lead the company's computer-vision platform: one config-driven system covering data ingestion, training-data preparation, model training, geospatial evaluation and GPU inference in production.
- Train modern deep-learning segmentation models for tree species and forest-damage mapping, from transformer-based architectures with foundation-model backbones to classic CNNs, across semantic, instance and panoptic tasks.
- Built self-supervised pretraining on unlabelled aerial imagery and semi-supervised pseudo-labelling to make the most of scarce annotations.
- Built the training-data foundation: a versioned data lake with provenance, intake of finished annotation jobs, reproducible dataset recipes and leakage-safe splits.
- Turn pixel predictions into GIS-ready maps: tiled inference over very large GeoTIFFs, vectorisation, geometry repair and certainty scoring.
- Contributed to the orthomosaic pipeline: LiDAR terrain models, direct georeferencing from drone metadata, seamline mosaicking and photogrammetric true orthophotos.
- Led the tree stem-diameter estimation project end to end, from field reference data and aerial height data to a production model with staged releases.
- As AI lead, set up an AI-agent-assisted engineering workflow and designed an AI-agent research process supporting the company's strategy work.

Jan – Jun 2021

Master's Thesis, Ericsson Research

Ericsson AI Research, with Linköping University

- Built a random-forest classifier that flags moments of high end-to-end latency in measurements from Ericsson's 5G test network.
- Used causal Bayesian networks (structure learning with the PC algorithm, then causal-effect estimation) to find which network measurements actually drive the latency, not just move with it.
- Thesis: *Anomaly detection and causal inference in 5G mobile networks*, Linköping University, 2021. Summary and report.

2005 – 2019

Seismic Data Acquisition QC Supervisor (junior to senior)

National Iranian South Oil Company (NISOC), Khuzestan, Iran

- Quality control of 2D and 3D land seismic acquisition in deserts, sand dunes and mountains: recording quality, geometry and field procedures.
- Supervised acquisition in more than five oil fields, with dynamite and vibroseis sources; reviewed survey design and field data before processing.
- Mentored junior geophysicists; recognised as one of the company's best seismic acquisition supervisors in 2015.

EDUCATION

2019 – 2021

MSc Statistics and Machine Learning

Linköping University, Sweden

Two-year programme (120 ECTS); thesis with Ericsson Research (see Experience).

2009 – 2012 **MSc Geophysics (Seismology)**
Institute of Geophysics, University of Tehran, Iran

Thesis: *Petroleum reservoir characterization using seismic data analysis* (prestack seismic inversion).

2000 – 2004 **BSc Petroleum Exploration Engineering**
Petroleum University of Technology, Ahvaz, Iran

SKILLS

Computer vision	Semantic, instance and panoptic segmentation; object detection; vision transformers and foundation-model backbones; self- and semi-supervised learning; OpenCV
Deep learning	PyTorch, Hugging Face Transformers, Detectron2, segmentation-models-pytorch, timm; TensorFlow/Keras
Geospatial	Rasterio, GDAL, GeoPandas, Shapely, PDAL; LiDAR; GeoTIFF/GeoPackage
Data & tooling	Python, NumPy, Polars, DuckDB, Pydantic, pytest, Streamlit, Dash, CVAT
Statistics	Regression, validation design, Bayesian methods, causal inference
Basic / working knowledge	Docker, AWS (S3, SQS, SageMaker), GitHub Actions; photogrammetry, orthorectification and remote sensing (learning); C++ (learning)